Example 2.12

A packet consists of 100 ball pens. The distribution of the number of defective ball pens in each packet is given below:

x	0	1	2	3	4	5
f	61	14	10	7	5	3

Examine whether Poisson distribution is appropriate for the above data at 5% level of significance.

Solution:

Step 1 : **Null hypothesis** H_0: Fitting of Poisson distribution is appropriate for the given data.

Alternative hypothesis H_1: Fitting of Poisson distribution is not appropriate for the given data.

Step 2 : **Data**

The expected frequencies are computed as under:

To find the mean of the distribution.

x	f	fx
0	61	0
1	14	14
2	10	20
3	7	21
4	5	20
5	3	15
Total	100	90

$$\bar{x} = \frac{\sum fx}{\sum f} = \frac{90}{100} = 0.9$$

Probability mass function of Poisson distribution is:

$$p(x) = \frac{e^{-m} m^x}{x!};\ x = 0, 1, \ldots \qquad (2.2)$$

In the case of Poisson distribution mean $(m) = \bar{x} = 0.9$.

At $x = 0$, equation (2.2) becomes

$$p(0) = \frac{e^{-m}\ m^0}{0!} = e^m = e^{0.9} = 0.4066 .$$

The expected frequency at x is $N\,P(x)$

12th_Statistics_EM_Unit_2.indd 65 1/28/2020 7:10:10 PM